AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-1411	**Classification:**	Critical / Elemental Impurities (ICH Q3D)
Product / API Name:	Azithromycin Dihydrate API (USP/EP)	**Batch / Lot Number:**	AZM-2026-640
Manufacturing Stage:	Stage 4: Palladium Scavenging & Crystallization	**Equipment Tag:**	Scavenger Column SCV-01 / Agilent 7900 ICP-MS
Date of Occurrence:	2026-10-13 19:40 UTC	**Date of Discovery:**	2026-10-14 17:00 UTC
Material / Lot ID:	Silica Thiol Scavenger Resin SIL-THI-304	**Reporting Unit:**	Trace Elemental Spectroscopy Lab

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
Elemental Impurity: Palladium (Pd)	**NMT 10.0 ppm (ICH Q3D Oral PDE)**	**18.2 ppm [OOS]**	ICP-MS with Microwave Digestion
Elemental Impurity: Nickel (Ni)	NMT 20.0 ppm	1.2 ppm	ICP-MS
Assay (Anhydrous Azithromycin)	94.5% - 103.0%	97.4%	HPLC Amperometric Detection
Crystallinity (Microscopy)	Birefringent needles / laths	Conforms	Polarizing Optical Microscopy

2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

ICP-MS screening of Azithromycin Dihydrate batch AZM-2026-640 revealed residual palladium catalyst at 18.2 ppm against the ICH Q3D limit of 10.0 ppm. Examination of metal scavenger column SCV-01 demonstrated preferential channeling and fluid bypass around the silica-thiol resin bed due to improper top frit distribution disc alignment during column re-packing on 2026-10-12.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. 950 kg of Azithromycin Dihydrate bulk quarantined in Warehouse 2.
2. Column SCV-01 decommissioned; standardized slurry packing SOP instituted.
3. Re-treatment with secondary thiol resin batch adsorption protocol under review.

Prepared by:	C. Lindqvist (Spectroscopist)	2026-10-14
Verified by:	K. Kowalski (Production Lead)	2026-10-14
QA Approval:	Dr. E. Gomez (QA Lead)	**Status: PENDING ROOT CAUSE ANALYSIS**